Pollard rho across the strata of an isogeny volcano

Question. Take one ECDLP instance, carry it by explicit isogenies to every level of an ℓ-isogeny volcano, and run Pollard rho on each curve: does the level (crater → floor), or the particular curve, change how fast rho solves it? Report min, max and mean.

Update: a fourth curve, D (j = 0 crater, `run-20261017T0140Z`), shows the one place the curve matters: there, rho folded by the order-6 automorphism needs 1.89 ± 0.09× fewer operations ($\sqrt{3}$ predicted).

Answer (measured on three curves: A, `run-20261016T2302Z`, $\ell = 3$, ramified crater; B, `run-20261016T2315Z`, $\ell = 3$, split crater; C, `run-20261017T0049Z`, $\ell = 2$, a whole depth-3 2-volcano). No measurable effect on any of them. Curve A in detail: All 29 isogenous curves and 3 isomorphic controls took the same work to within sampling noise. Pooled over the 2,900 isogenous solves: min 8,477, mean 659,263 ± 12,422 (95 % CI), max 2,099,707 group operations, against $\sqrt{(\pi n/2)} = 654{,}547$. Every level's 95 % CI includes or touches that value, and per-step cost is flat (641–661 ns/op on every curve). That is what theory predicts: isogenous curves over F_p have the same group order, and generic rho depends only on the order of $\langle P \rangle$. The spread *between* curves (per-curve means 572,846 – 733,141) matches the spread of 100-trial means of a quantity whose single-solve sd is about 341,000 (sd of a mean ≈ 34,000).

Method

`volcano-rho` (`src/bin/volcano_rho.rs`, native Rust, all statistics computed there; `plot.py` only lays out recorded values):

1. **Curve search.** Random curves $y^2 = x^3 + ax + b$ over random 40-bit primes $p \equiv 3 \pmod 4$, point count by baby-step giant-step, until $D = t^2 - 4p = f^2 \cdot D_K$ has $v_3(f) = 3$ (a 3-volcano with three levels below the crater) and $\#E = h \cdot n$ with n prime, $h \le 64$.
2. **Walk.** Breadth-first over 3-isogenies. Each edge's kernel is a rational root x_0 of the 3-division polynomial $\psi_3 = 3x^4 + 6ax^2 + 12bx - a^2$, and the map is Vélu's, applied explicitly to points. Vertices are keyed by j-invariant. A vertex's level is $\mathrm{depth} - \mathrm{dist}(v, \mathrm{floor})$, the floor being the vertices with exactly one rational 3-kernel; the tool asserts that no edge skips a level.
3. **One problem everywhere.** P of order n and $Q = kP$ on the root curve are pushed along the BFS tree to every curve. Each edge is checked: images lie on the codomain, $\varphi(P+Q) = \varphi(P) + \varphi(Q)$, and $n \cdot \varphi(P) = O$ with $\varphi(P) \neq O$ ($3 \nmid n$, so every map is injective on $\langle P \rangle$). Every rho solve on every curve returned the same k.
4. **Rho.** Single walker, 32-adding walk (Teske), distinguished points with $\theta = 2^{-8}$, no negation map; restarts on a DP-free run longer than $40/\theta$. Trial seed s uses the same walk constants on every curve (the partition hashes x, so walks differ). Measured unit: group additions to the solve; also wall time per solve.
5. **Control.** Three random isomorphic re-scalings of the root, $(u^4 a, u^6 b)$, $(x, y) \mapsto (u^2 x, u^3 y)$: the same curve in new coordinates, showing the spread that a pure change of representation produces.
6. All 3,200 (arm, seed) jobs shuffled and run on 4 threads.

Curve A: the instance (`run-20261016T2302Z`, ramified crater)

p	1090993606999 (40 bits)
root E	a = 690282649366, b = 1050606678233
#E	1090993806052 = 4 · n, n = 272748451513 (prime, 38 bits)
trace t	−199052
$D = t^2 - 4p$	$-4324352729292 = 54^2 \cdot (-1482974187)$; $v_3(f) = 3$
crater	$D_K \equiv 0 \pmod 3$: ramified, 2 crater vertices joined by one edge
k	49631396114 (Q = kP)
volcano	80 curves = 2 + 6 + 18 + 54 (crater, L1, L2, floor), 79 edges, all verified

Curve A results (iterations = group operations; time on a shared

4-vCPU VM) Pooled by level:

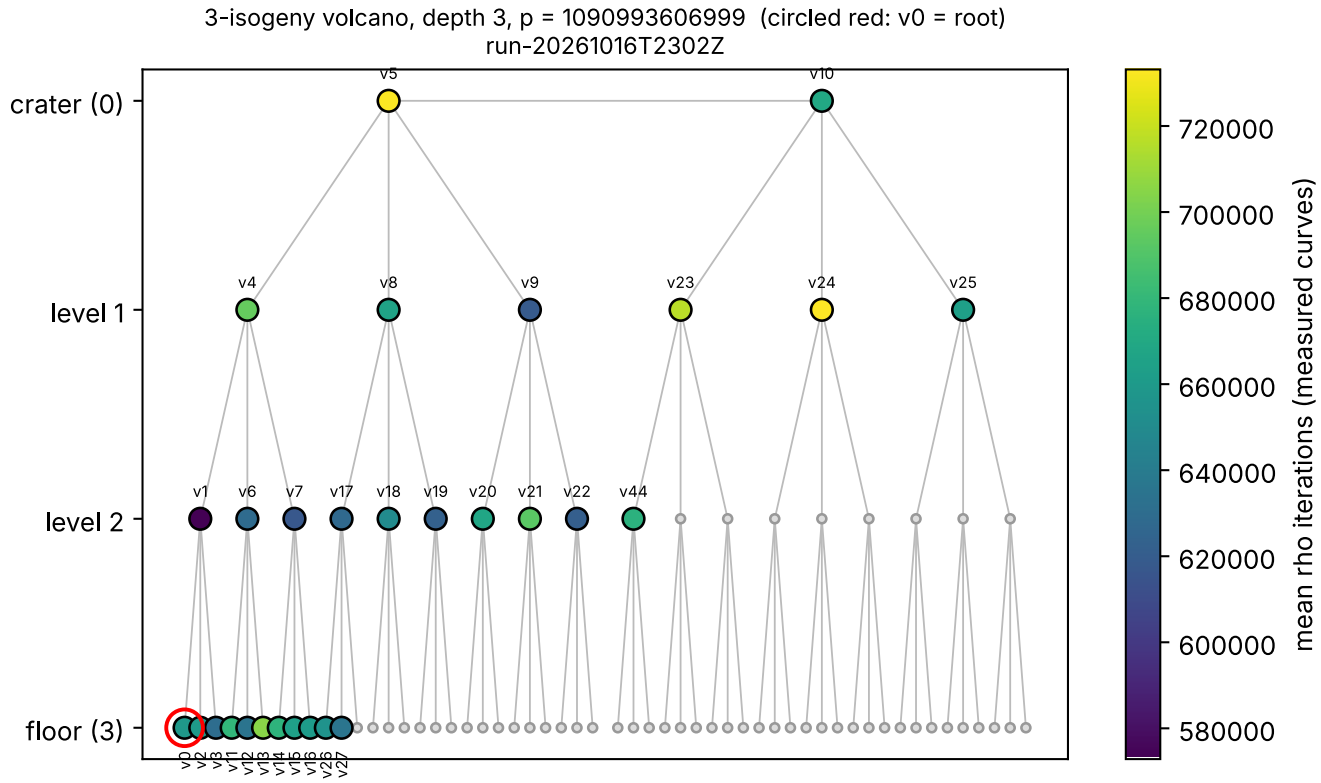
level	curves	solves	min	mean ± 95 % CI	median	max	mean ms ± CI
0 crater	2	200	17,415	700,483 ± 47,849	662,463	1,621,252	456.9 ± 32.0
1	6	600	48,186	681,766 ± 26,525	656,746	1,861,764	440.7 ± 17.6
2	10	1,000	9,649	637,158 ± 20,410	598,424	1,792,889	414.4 ± 13.6
3 floor	11	1,100	8,477	659,589 ± 20,993	620,458	2,099,707	426.0 ± 13.9
all isogenous	29	2,900	8,477	659,263 ± 12,422	622,386	2,099,707	427.1 ± 8.2
isomorphic control	3	300	23,698	674,042 ± 40,816	628,245	1,820,972	434.0 ± 27.0

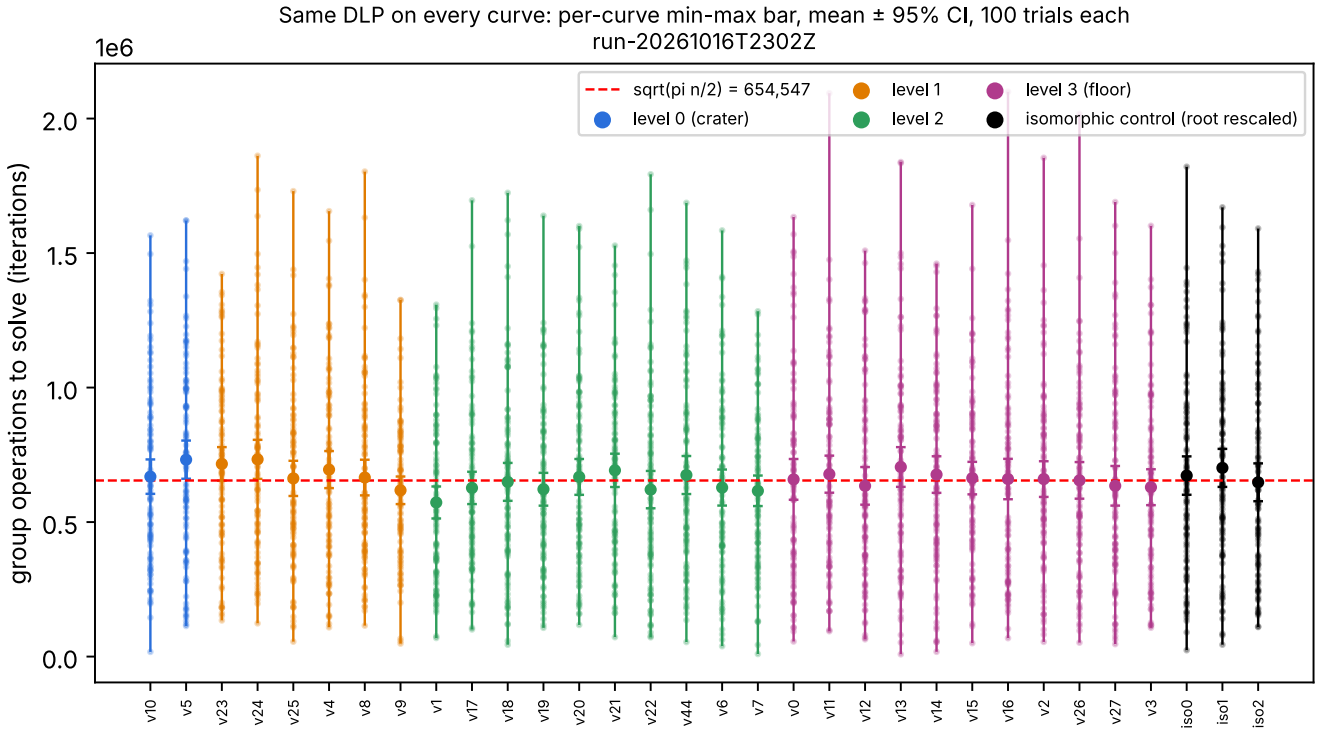
Reference: $\sqrt{(\pi n/2)} = 654{,}547$ (+ about 2^8 for the distinguished-point tail).

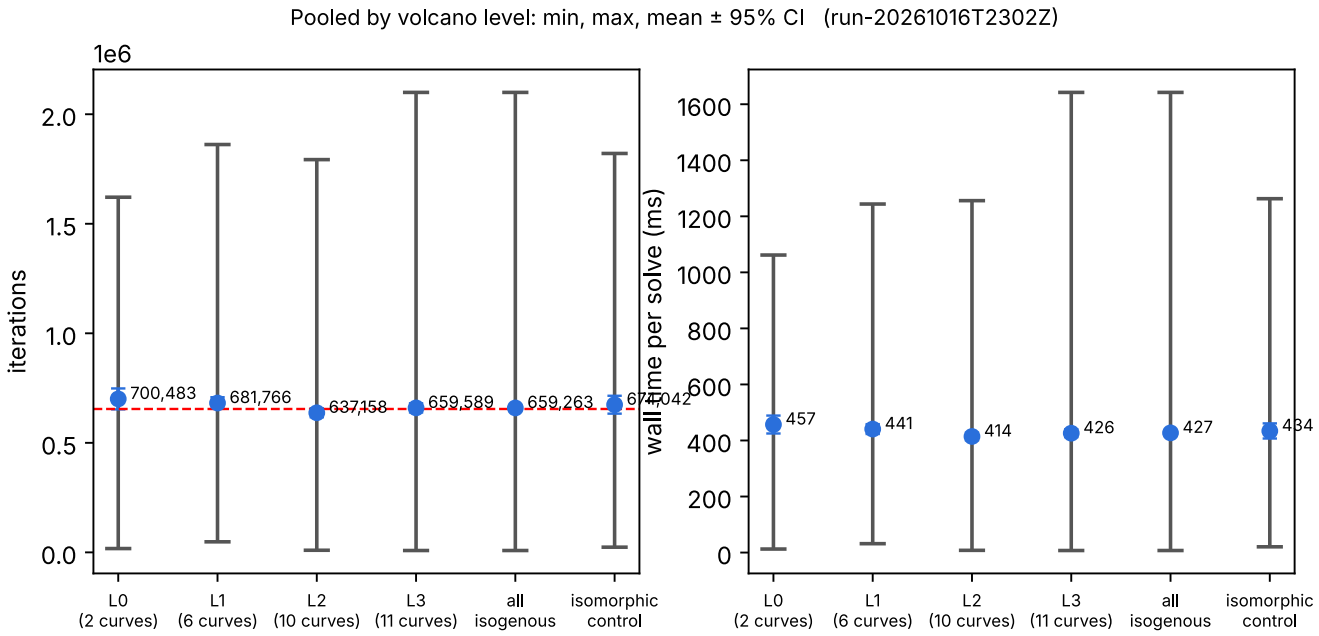
Per curve (100 solves each), iterations:

curve	level	min	mean	max		curve	level	min	mean	max
v5	0	113,746	732,044	1,621,252		v18	2	44,056	649,444	1,724,073
v10	0	17,415	668,921	1,565,882		v19	2	106,910	621,995	1,638,981
v4	1	108,640	695,082	1,655,814		v20	2	117,650	667,890	1,600,331
v8	1	115,038	665,553	1,802,980		v21	2	72,048	692,313	1,527,838
v9	1	48,186	618,080	1,325,879		v22	2	71,551	620,647	1,792,889
v23	1	134,281	716,324	1,422,523		v44	2	54,056	674,911	1,686,754
v24	1	123,525	733,141	1,861,764		v0 (root)	3	55,979	658,723	1,634,032
v25	1	55,435	662,413	1,730,218		v2	3	54,631	659,817	1,854,099
v1	2	69,112	572,846	1,308,207		v3	3	107,168	629,575	1,601,367
v6	2	38,512	628,345	1,584,239		v11	3	94,240	678,053	2,094,241
v7	2	9,649	616,240	1,283,603		v12	3	64,211	634,612	1,509,154
v17	2	100,833	626,950	1,696,010		v13	3	8,477	704,738	1,838,374
iso0	ctrl	23,698	672,888	1,820,972		v14	3	17,758	676,654	1,460,561
iso1	ctrl	44,112	701,414	1,670,056		v15	3	49,801	663,488	1,678,869
iso2	ctrl	109,711	647,822	1,591,405		v16	3	68,503	659,946	2,099,707
						v26	3	51,815	655,015	2,017,181
						v27	3	46,289	634,861	1,689,685

Full per-curve rows (with times, ns/op, medians, p10/p90) are in `runs/run-20261016T2302Z/results.json`; every solve is a row of `trials.csv`.







Curve B: a second volcano with a split crater (`run-20261016T2315Z`)

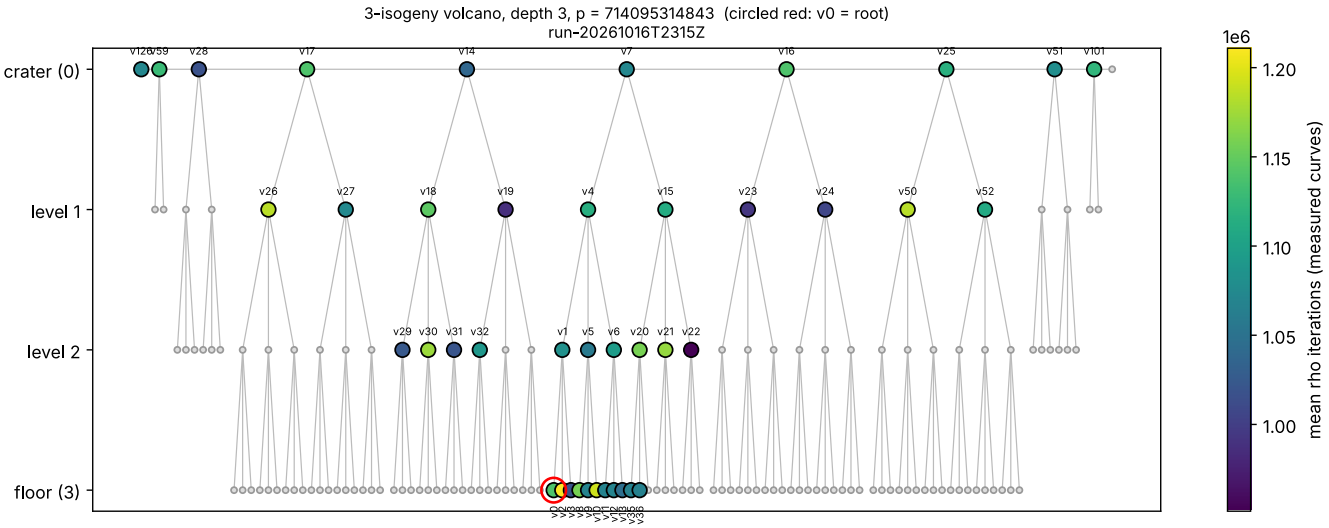
Same tool, `--crater split --seed 20261017`. This volcano's crater is a cycle, so the walk reaches a long run of crater vertices, each with its own depth-3 tree. The cycle is truncated at the walk radius.

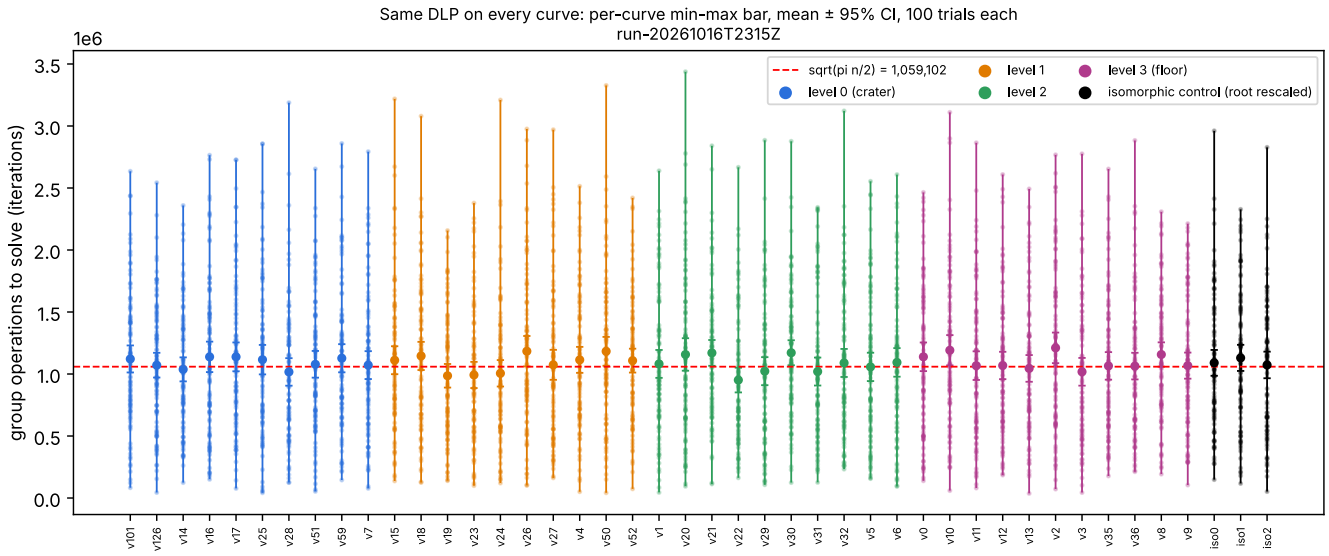
p	714095314843 (40 bits)
root E	a = 354455170800, b = 66680642576
#E	714094928137 = n, prime (cofactor 1)
trace t	386707
D = t² − 4p	−2706838955523 = 27² · (−3713084987); $v_3(f) = 3$
crater	D_K ≡ 1 (mod 3): split, the crater is a cycle
k	661235192333
walk	323 curves, 322 edges verified; inside the measured radius 11 crater, 18, 42 and 90 floor curves by level

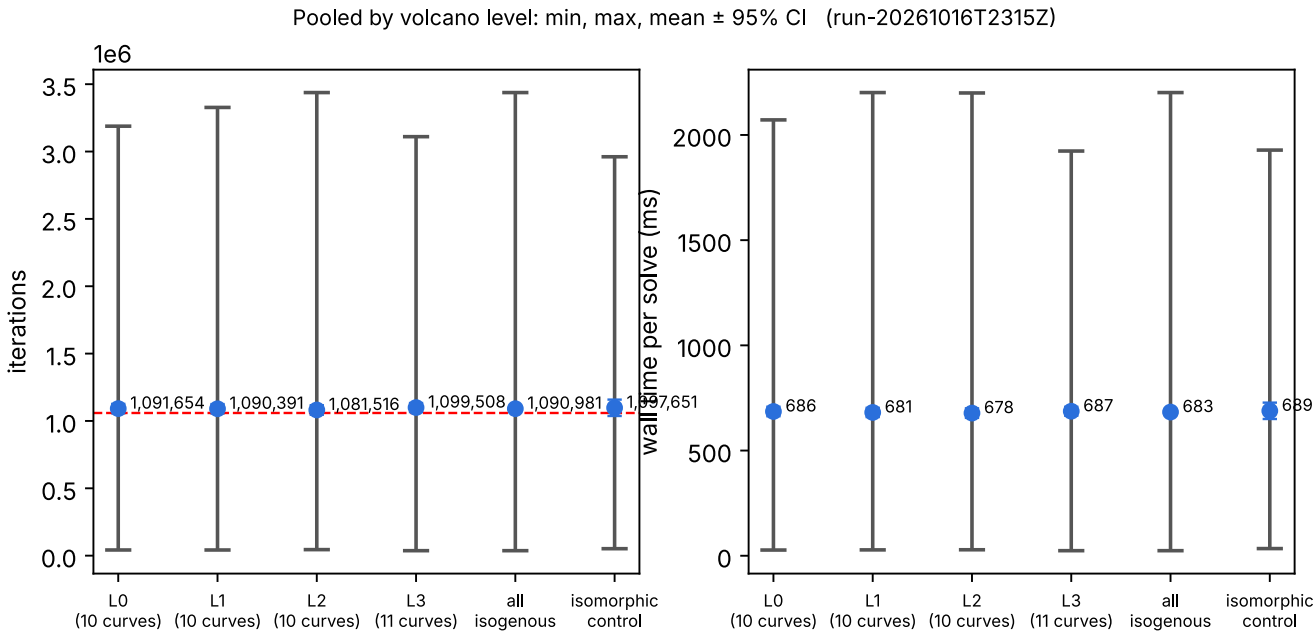
rho	41 curves + 3 controls × 100 solves = 4,400, all correct; 754 s

level	curves	solves	min	mean ± 95 % CI	median	max	mean ms ± CI
0 crater	10	1,000	42,088	1,091,654 ± 35,041	1,032,925	3,188,276	685.7 ± 22.6
1	10	1,000	42,018	1,090,391 ± 34,841	1,046,357	3,327,413	681.2 ± 22.0
2	10	1,000	44,771	1,081,516 ± 35,498	1,012,223	3,437,779	678.1 ± 22.7
3 floor	11	1,100	36,930	1,099,508 ± 33,756	1,050,592	3,110,345	686.8 ± 21.1
all isogenous	41	4,100	36,930	1,090,981 ± 17,380	1,034,403	3,437,779	683.0 ± 11.0
isomorphic control	3	300	51,690	1,097,651 ± 60,787	1,094,657	2,961,196	688.9 ± 38.9

√(πn/2) = 1,059,102. Per-curve means span 951,361 (v22) to 1,210,896 (v2); the per-op cost is 622–635 ns on every curve. Per-curve rows are in `runs/run-20261016T2315Z/results.json`.







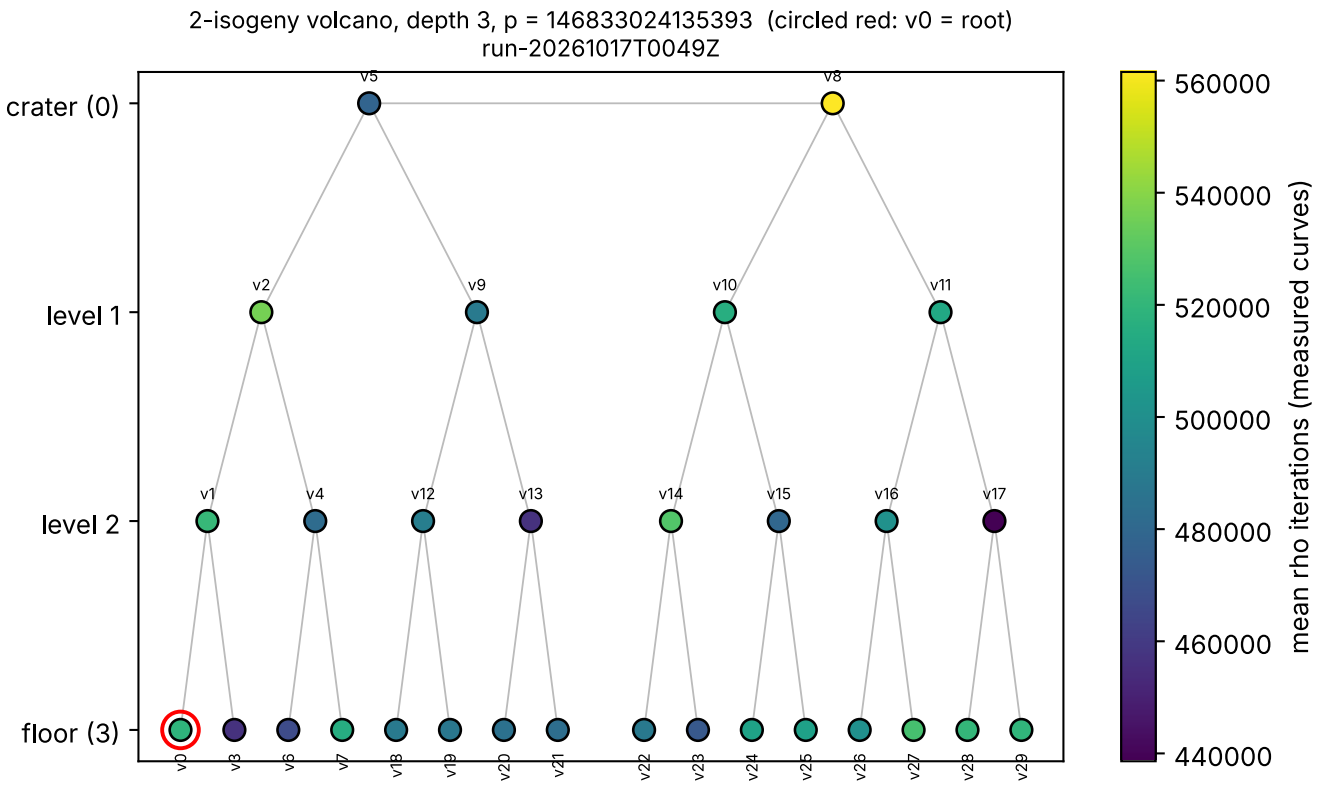
Curve C: a whole 2-isogeny volcano (`run-20261017T0049Z`)

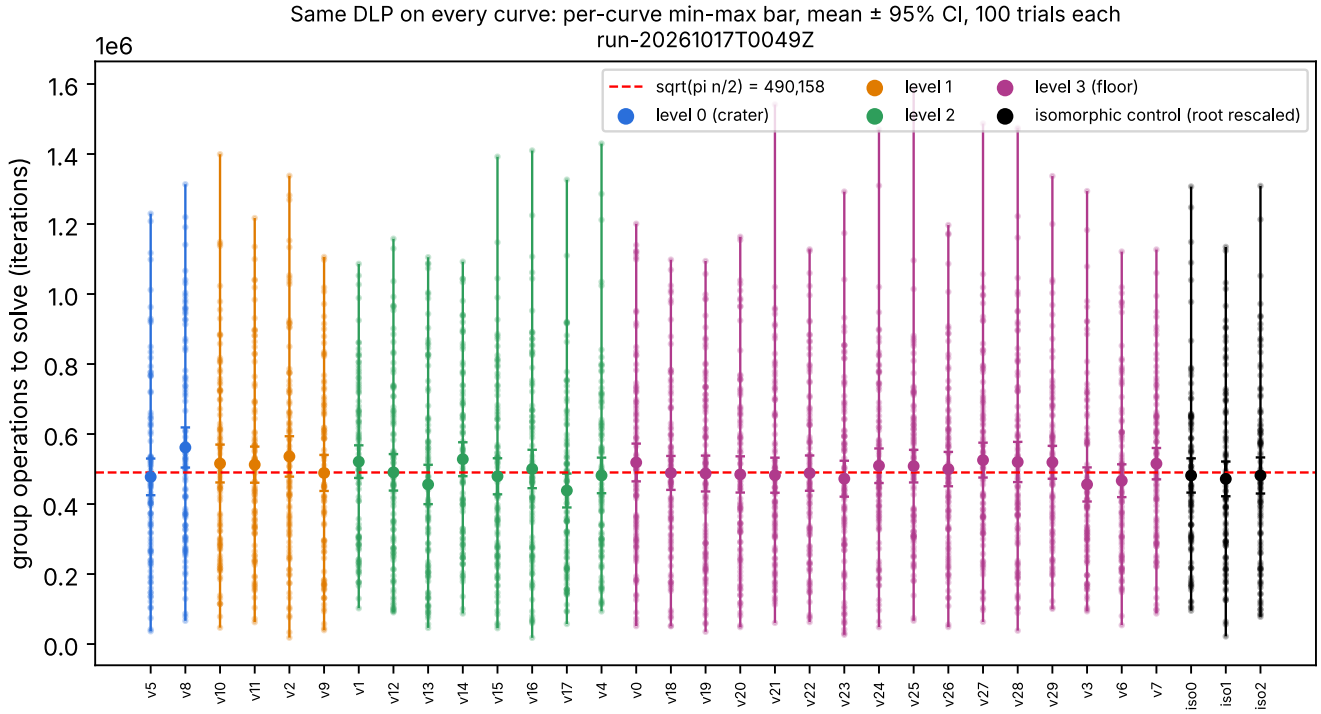
Same tool with 2-isogenies: `--ell 2 --bits 48 --depth 3 --max-cofactor 4096 --seed 20261018`. A 2-volcano deeper than one level needs $p \equiv 1 \pmod 8$; with $p \equiv 3 \pmod 4$, $v_2(t^2 - 4p) \le 3$. $\ell = 2$ searches draw such primes. Depth 3 also puts at least 2^6 into #E, hence the larger cofactor bound. The walk reaches the whole volcano, a binary tree under a two-vertex crater, and every curve in it is measured.

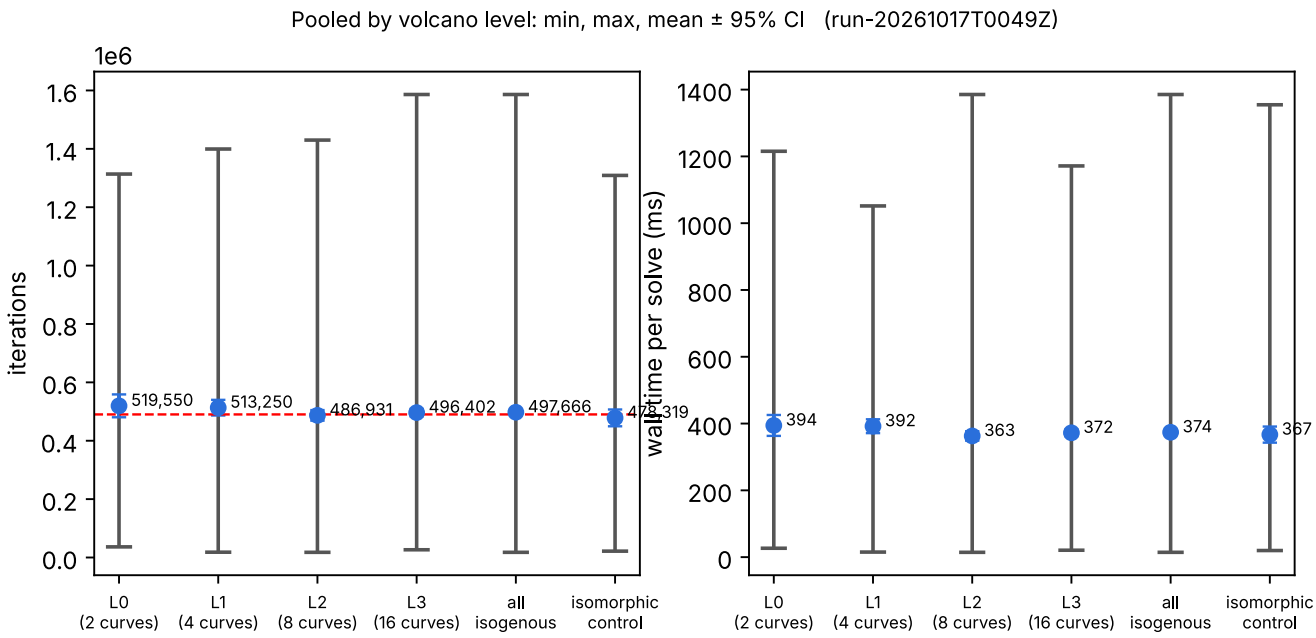
p	146833024135393 (48 bits, ≡ 1 mod 8)
root E	a = 14391545271884, b = 141956518099848
#E	146833030777920 = 960 · n, n = 152951073727 (prime, 37.2 bits)
trace t	−6642526
D = t² − 4p	−543208944880896 = 56² · (−173217138036); $v_2(f) = 3$
crater	ramified (D_K even), 2 crater vertices joined by one edge
k	50935020165
walk	30 curves = 2 + 4 + 8 + 16 (the whole volcano), 29 edges verified
rho	30 curves + 3 controls × 100 solves = 3,300, all correct; 310.7 s
cross-check	`ca volcano --ell 2` (aburan28/cryptanalysis) counts the same #E and finds the same 30 curves, 2/4/8/16

level	curves	solves	min	mean ± 95 % CI	median	max	mean ms ± CI
0 crater	2	200	36,433	519,550 ± 39,163	479,542	1,313,611	394.4 ± 31.3
1	4	400	18,197	513,250 ± 26,834	487,539	1,399,367	392.0 ± 20.7
2	8	800	17,768	486,931 ± 18,140	465,983	1,429,985	362.9 ± 14.1
3 floor	16	1,600	26,502	496,402 ± 12,454	466,178	1,586,151	372.3 ± 9.7
all isogenous	30	3,000	17,768	497,666 ± 9,336	467,154	1,586,151	373.9 ± 7.3
isomorphic control	3	300	21,764	478,319 ± 28,790	447,625	1,308,949	366.9 ± 24.1

$\sqrt{(\pi n/2)}$ = 490,158 (×1.016 for the r = 32 walk: 498,000). Per-curve means span 438,599 (v17) to 561,536 (v8). Per-op cost is 736–800 ns on every curve. Every level's CI contains $\sqrt{(\pi n/2)}$ or its r-adding value, and every pair of level CIs overlaps.


2-isogeny volcano, depth 3, p = 146833024135393 (circled red: v0 = root)
run-20261017T0049Z
crater (0)
level 1
level 2
floor (3)
v5
v8
v2
v9
v10
v11
v1
v4
v12
v13
v14
v15
v16
v17
v0
v3
v6
v7
v18
v19
v20
v21
v22
v23
v24
v25
v26
v27
v28
v29
mean rho iterations (measured curves)
560000
540000
520000
500000
480000
460000
440000


Same DLP on every curve: per-curve min-max bar, mean ± 95% CI, 100 trials each
run-20261017T0049Z
1e6
group operations to solve (iterations)
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
sqrt(pi n/2) = 490,158
level 0 (crater)
level 1
level 2
level 3 (floor)
isomorphic control (root rescaled)
v5
v8
v10
v11
v2
v9
v1
v12
v13
v14
v15
v16
v17
v4
v0
v18
v19
v20
v21
v22
v23
v24
v25
v26
v27
v28
v29
v3
v6
v7
iso0
iso1
iso2



Why min and max differ so much

Pollard rho is randomized: it walks pseudo-randomly through ⟨P⟩ until the walk meets a point it has visited before. The step at which that happens is a birthday-problem collision time: `P(no collision after k steps) ≈ exp(−k²/2n)`. That gives a mean of `√(πn/2) ≈ 1.25√n` and a standard deviation of `√((4 − π)n/2) ≈ 0.66√n`, 52 % of the mean. The spread between min and max is that distribution's width, not anything about a curve.

Observed vs the birthday model (derived from the recorded `trials.csv`, all isogenous solves):

curve	solves	mean (model)	sd (model)	1st pct (model)	median (model)	99th pct (model)
A	2,900	659,263 (654,547)	341,296 (342,147)	71,551 (74,043)	622,025 (614,906)	1,584,239 (1,584,962)
B	4,100	1,090,981 (1,059,102)	567,779 (553,617)	124,423 (119,807)	1,034,344 (994,960)	2,609,159 (2,564,577)
C	3,000	497,666 (490,158)	260,902 (256,217)	59,046 (55,447)	467,109 (460,473)	1,190,999 (1,186,900)

- **One curve, 100 solves:** the model's typical fastest solve is `√(2n·ln(101/100))` and its typical slowest `√(2n·ln 101)`, about 0.14√n and 3.0√n. That is a 20× spread on every curve, the controls included.
- **Pooled extremes:** pools of 1,000+ solves reach further into both tails, which is where single minima like 8,477 come from.
- **Per-curve means:** each mean averages 100 such draws, so it carries about `0.66√n / 10` of noise: model 34,215 / 55,362 / 25,622 for A / B / C. The measured spread of the per-curve means is 36,686 / 62,351 / 26,723, so the curves differ by sampling noise alone.
- **Why the curve doesn't matter:** an isogenous curve changes the coordinates, so the x-hash that drives the walk takes different paths. The group has the same order, so the collision-time distribution is the same.
- **What a real effect would need:** a curve effect would have to move the mean by more than this noise. Detecting an x % difference takes roughly `(2 · 0.52 · 100/x)²` solves per arm, about 1,100 for 3 %.

Same walk on every curve: the pull-back control

(`run-20261017T0111Z`) The per-curve differences come from the walk, not the curve. Rho picks each step, decides distinguished points and keys its table by a hash of the current point's x-coordinate. An isogeny or a rescaling changes those coordinates, so each curve runs a different pseudo-random walk on the same group.

Prediction. If every curve read the root curve's coordinates instead, the walks would coincide and every curve would take exactly the same number of steps for each seed.

How the walk reads the root's coordinates (`--walk pullback`).

- **The canonical point.** Each curve m isogeny steps from the root

reads `[ℓ^(M−m)] · φ̂₁ ∘ … ∘ φ̂_m (R)` on the root curve, which equals `±ℓ^M φ⁻¹(R)`. Here M is the largest distance among the measured curves, 7 for curve C. - **The dual edge.** Each dual $\hat{\varphi}$ is the rational kernel whose codomain has the parent's j-invariant, composed with the isomorphism onto the parent's model. It is accepted only if `φ(φ̂(P)) = ±[ℓ]P`. - **Controls.** These pull back by undoing their rescaling.

Result on curve C: 30 curves + 3 controls × 20 seeds.

	result
seeds that took the same number of steps on all 33 arms	**20 / 20**

all 660 solves	correct
steps per seed (sorted)	67,490 · 165,318 · 202,642 · 212,430 · 226,841 · 274,404 · 369,316 · 497,854 · 524,385 · 593,906 · 621,975 · 624,229 · 722,575 · 737,697 · 745,537 · 755,347 · 767,003 · 839,832 · 903,373 · 1,021,850
mean over seeds	543,700 (√(πn/2) = 490,158; one draw per seed, sd of the mean ≈ 57,000)
cost per step	6,425 ns (root, chain length 0, ℓ^7 lift) rising to 7,562 ns (chain length 7); native walk 757 ns

The native walk on the same curves gives per-seed counts that are uncorrelated across curves (mean correlation +0.002 on A, −0.025 on C). The pulled-back walk gives the same count everywhere. The step count is a property of the walk and the instance; the curve only supplies the coordinates the walk hashes. Pulling back makes every step about 9× dearer and saves no steps: whichever walk a curve runs, its collision time follows the same birthday distribution.

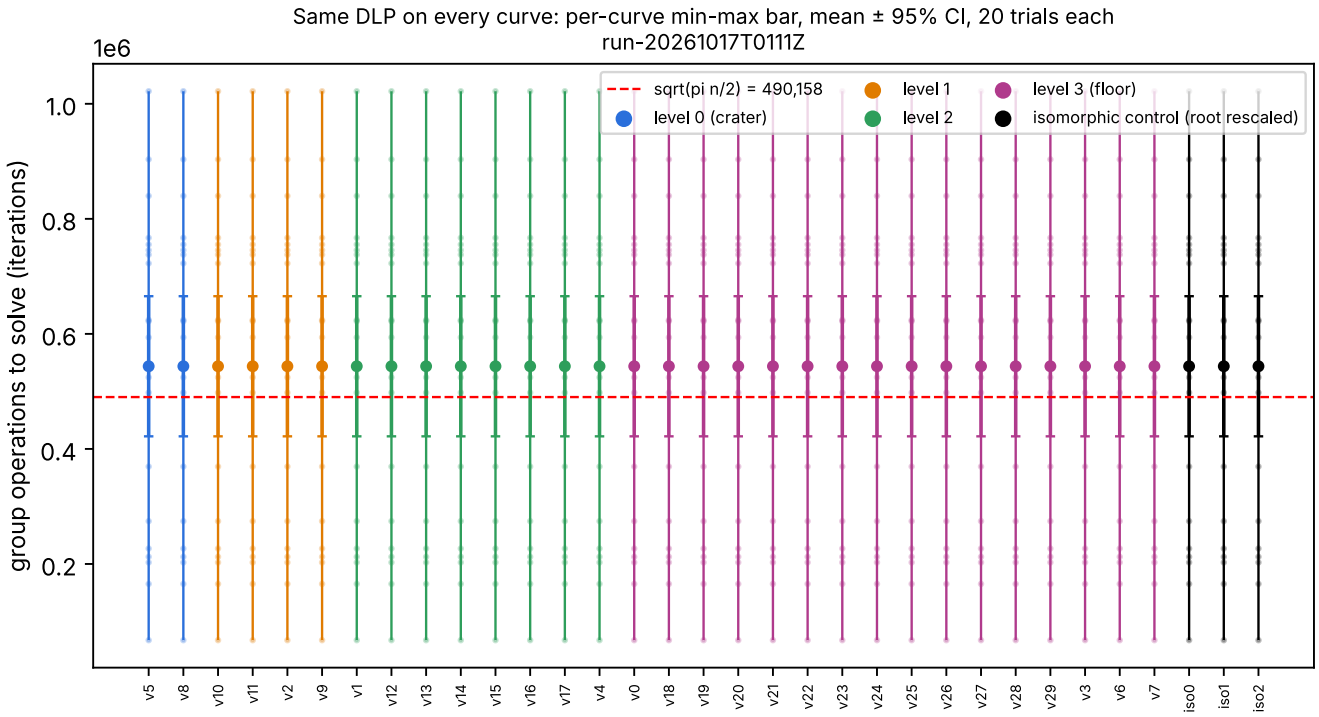


Figure 1: pull-back per curve

Where the curve does matter: the j = 0 crater (`run-20261017T0140Z`)

On a prime field, the one property that can differ between isogenous curves and change rho's step count is the automorphism group. A curve with j = 0 has six automorphisms; every other curve has two (±1). Rho can walk on classes of six points instead of two, which predicts √(6/2) = √3 ≈ 1.73× fewer steps than the negation-map walk. Such curves exist only in classes with D_K = −3, where j = 0 sits on the crater. For a D_K = −3 class with a 2-volcano, (−3/2) = −1, so the crater is the single j = 0 vertex, and its three 2-isogenies are permuted by the order-3 automorphism and land on one curve.

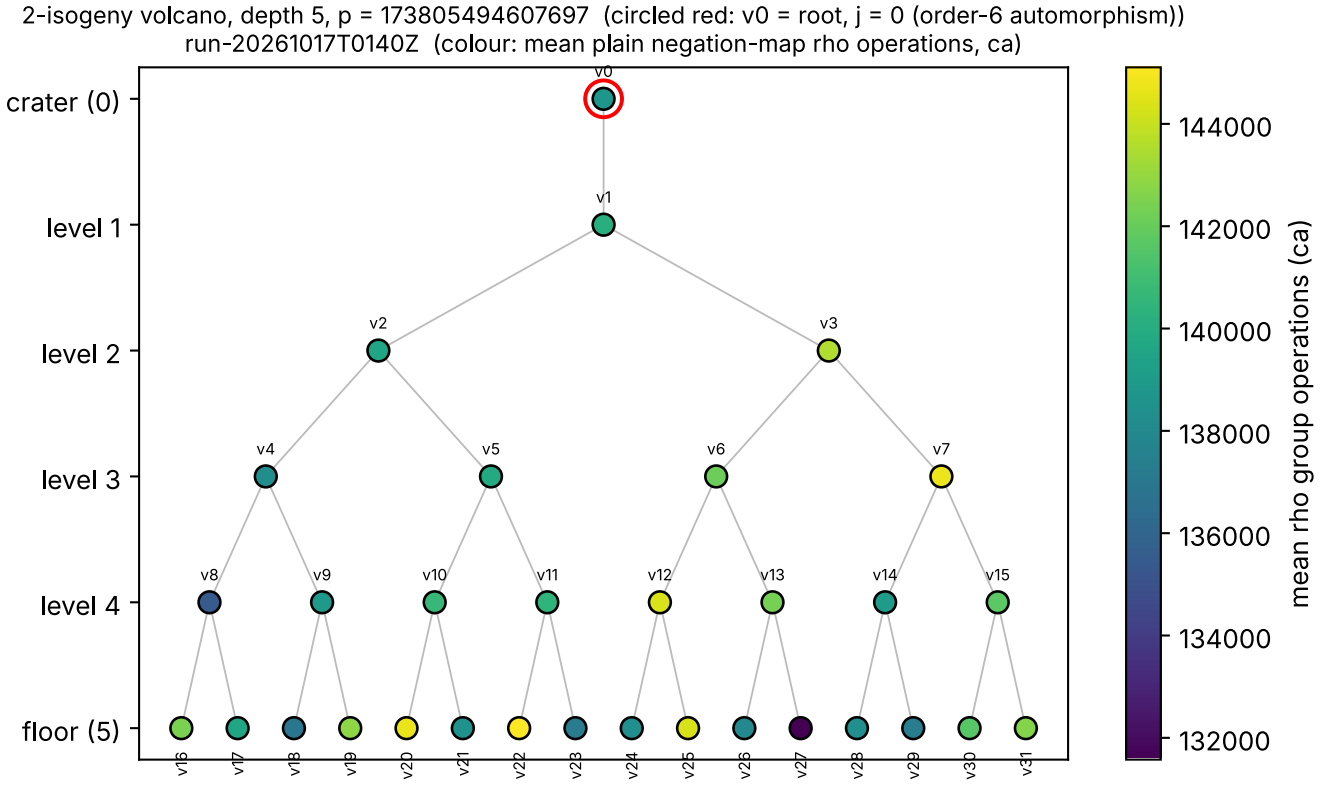
Curve D. It was found by `volcano-rho --j0 --ell 2 --bits 48 --depth 5 --max-cofactor 100000 --seed 20261015`, which searches $y^2 = x^3 + b$ with $p \equiv 1 \pmod{24}$. It was measured by `ca volcano … --automorphisms` from aburan28/cryptanalysis at `e4f9e25`; exact commands are in `runs/run-20261017T0140Z/command.txt`. Every curve runs `ca`'s plain negation-map rho. The j = 0 crater curve also runs `ca_curve_solve`'s automorphism-folded rho on the same instance with the same seeds.

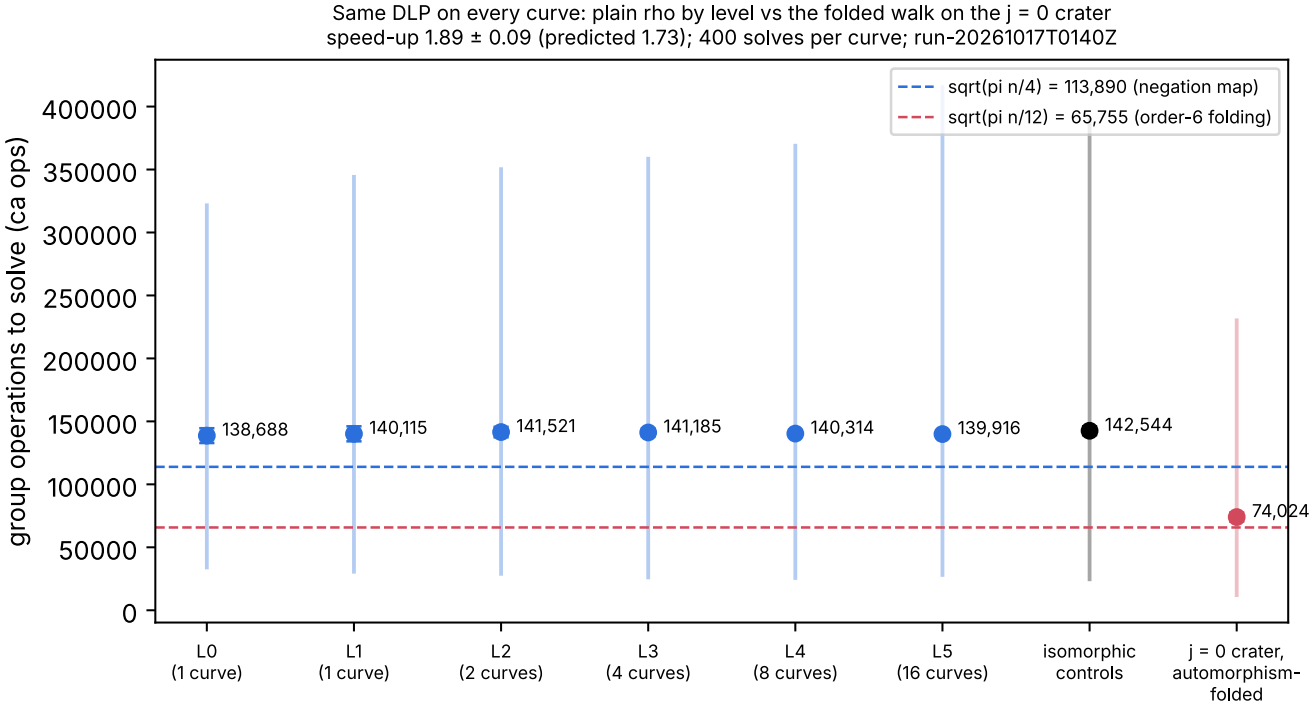
p	173805494607697 (48 bits, ≡ 1 mod 24)
root E	$y^2 = x^3 + 26953713445357$, j = 0
#E	173805517224804 = 10524 · n, n = 16515157471 (prime, 33.9 bits)
trace t	−22617106; D_K = −3, 2-volcano depth 5, inert crater
walk	32 curves = 1 + 1 + 2 + 4 + 8 + 16, 31 edges verified (the whole volcano)
rho	32 curves + 3 controls (plain) + the crater curve (folded), 400 seeds each: 14,400 solves, all correct; 124.0 s

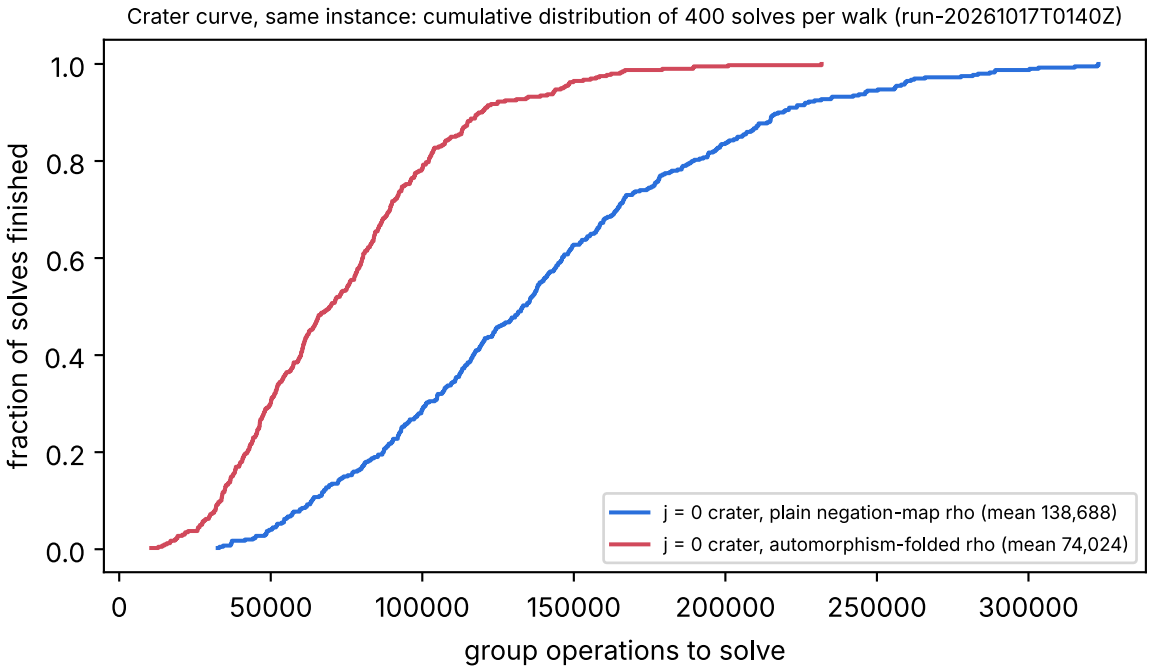
arm	curves	solves	min	mean ± 95 % CI	max

L0, the j = 0 crater, plain	1	400	32,500	138,688 ± 5,881	323,112
L1	1	400	29,091	140,115 ± 5,964	345,705
L2	2	800	27,442	141,521 ± 4,151	351,803
L3	4	1,600	24,572	141,185 ± 2,972	360,062
L4	8	3,200	24,150	140,314 ± 2,092	370,427
L5, the floor	16	6,400	26,545	139,916 ± 1,498	416,873
all isogenous, plain	32	12,800	24,150	140,242 ± 1,053	416,873
isomorphic controls, plain	3	1,200	23,098	142,544 ± 3,447	390,953
j = 0 crater, automorphism-folded	1	400	10,573	**74,024 ± 3,560**	231,754

- **Operations (measured):** the folded walk needs **1.89 ± 0.09**× fewer group operations than the plain walks, 140,242 against 74,024. On the crater curve alone, same instance and seeds, it's 138,688 against 74,024 (1.87×). Every plain level is flat at about 140k, as on curves A to C.
- **Against the √3 prediction:** the interval excludes 1.73. Both arms sit above their ideal values: plain at 1.23× $\sqrt{(\pi n/4)}$ = 113,890, folded at 1.13× $\sqrt{(\pi n/12)}$ = 65,755. The extra is `ca`'s walk overhead (its r = 1024 negation walk with fruitless-cycle handling), which falls more on the plain walk. The algorithmic factor is √3; this implementation's measured factor is 1.89.
- **Wall time (shared host, indicative):** 6.52 ms against 8.65 ms per solve, **1.33**×. A folded step costs 88 ns against 62 ns, because it picks a canonical representative among six points.
- **What this means for the volcano (derived):** an instance on any of the 31 lower curves can be carried up to the j = 0 crater by at most five 2-isogeny evaluations, which costs nothing next to 10^5 group operations, and solved there about 1.9× cheaper in operations (1.3× in time). This is the one place in a prime-field volcano where the curves are not interchangeable. It exists only for D_K = −3 (or −4, j = 1728: 2× fewer, √2 beyond negation) classes, and a random curve's class has |D_K| ≈ p, so there is nothing to climb to.
- **A recorded-output note:** the `ca` build that ran wrote the folded arm's mean into the crater vertex of `volcano.diagram.json`, giving a duplicate key. `plot_ca.py` takes vertex values from `results.json`'s plain arms, and `ca` is fixed (`31b72d2`).







Reading it

- **Observed:** level means differ by at most 63,000 iterations (L0 vs L2), and the 95 % CIs of L0 and L2 overlap. The CIs of L0, L2 and L3 contain $\sqrt{(\pi n/2)}$; L1's lower bound (655,241) sits 700 above it, with four level intervals and no correction for multiple comparisons. Single solves range over 250× (8,477 to 2,099,707) on every level and on the control, so min and max are the extremes of a wide distribution and say little about a curve. The mean is the meaningful statistic.
- **Curve B** (observed): the level means lie within 1.7 % of each other (1,081,516 to 1,099,508), and every level's CI overlaps every other's. All of them, and the control, sit about 3 % above $\sqrt{(\pi n/2)}$. The pooled mean, 1,090,981 ± 17,380, excludes it. Curve A's pooled mean sits 0.7 % above (inside its CI). A known heuristic (derived, not measured here) is that an r-adding walk needs about $1/\sqrt{(1 - 1/r)}$ times a truly random walk's steps: ×1.016 for r = 32, or 1,076,000 on B and 665,000 on A. Both pooled CIs contain those values. The offset is a property of the walk, shared by every curve and the control, not of any stratum.
- **Interpretation (derived, not measured):** an ℓ-isogeny with $\ell \nmid n$ is a group isomorphism $\langle P\rangle \longrightarrow \langle \varphi(P)\rangle$, so generic rho sees the same group of order n on every stratum; what changes between levels is the endomorphism ring, which generic rho does not use. The per-curve differences are what re-drawing the walk's random partition gives, as the isomorphic control (same curve, new coordinates) shows. Walking the volcano is not a way to make this generic DLP cheaper.
- **Gap to matched rho:** this *is* rho on every arm; no index-calculus arm is involved, so there is no IC baseline to improve and no §5a band to rank across.

Limits

- One instance per curve, three curves:
  - A: 40-bit p, ℓ = 3, depth 3, ramified crater with 2 vertices, so its L0 has only 200 solves;
  - B: 40-bit p, ℓ = 3, depth 3, split crater, truncated at the walk radius;
  - C: 48-bit p, ℓ = 2, depth 3, ramified crater, the whole volcano.
- Not an ecbench session: the counted unit is this tool's group-addition count; no isolation (shared 4-vCPU VM, 4 threads, jobs interleaved), so wall times are indicative only. Per-op time includes affine inversion by `i128` extended Euclid.
- Curves measured: A 29 of 80 (the whole volcano), B 41 of the 323 walked (root plus up to 10 per level in BFS order), C all 30.
- No negation map and no endomorphism-accelerated rho; a curve at the crater with a larger endomorphism ring would only help an algorithm that uses that ring (GLV-style speedups exist only for tiny |D_K|, and this D_K = −1482974187 is not tiny).

Run it on your own curve

```
cargo build --release --bin volcano-rho
target/release/volcano-rho --curve P,A,B [--order N] [--ell 3] [--seed S] [--trials 100] --out OUT
target/release/volcano-rho --curve P,A,B --engine ca --ca /path/to/ca ...   # cryptanalysis's C rho
python research/isogeny-volcano-rho/plot.py OUT    # optional: the standard graphs
```

The cryptanalysis repository has the same measurement built into its own binary, with no Rust involved: `ca volcano --p P --a A --b B [--trials T] [--diagram FILE]` (`ca_volcano.h`). It reproduces this study's curve A volcano exactly (80 curves, 2 / 6 / 18 / 54).

`y² = x³ + Ax + B` over `F_P`, any prime `P < 2^62`. Without `--order` the tool counts points (baby-step giant-step). It checks the order against random points, takes `n` the largest prime factor, draws a generator (cofactor × a seeded random point) and a secret `k` from `--seed`, and reads the volcano depth `v_ℓ(f)` and the crater shape from `D = t² − 4p`. Then it walks, carries `(P, kP)` to every curve and measures. A curve whose conductor ℓ does not divide has depth 0, so the walk covers crater curves only. Outputs go to `--out`: `results.json`, `trials.csv`, `instances.jsonl` (every curve's instance in `ca` format, ready for `ca solve`) and the standard graphs.

- **Engines.** `native` is this binary's single-walker 32-adding walk with distinguished points. The DP rate defaults to `min(8, log₂n/2 − 6)` bits, so small groups keep every point. With every point distinguished, the x-keyed table also catches ±X matches, so expect about √(πn/4). `--engine ca` runs `ca solve --alg rho` from aburan28/cryptanalysis on each curve, with its defaults (negation map on) plus any `--ca-arg`. Iterations are then `ca`'s reported `ops` and times its reported `seconds`.
- **Cost.** About √n group operations per solve: n ≈ 2^40 takes a second or so per solve on the native engine. Choose `--trials` with that in mind.
- **Determinism.** The same command and seed give the same curves, instance and iteration counts; wall times vary.

Reproduce

```
cargo build --release --bin volcano-rho
./target/release/volcano-rho --bits 40 --ell 3 --depth 3 --seed 20261016 \
    --trials 100 --per-level 10 --controls 3 --out OUT
./target/release/volcano-rho --bits 40 --ell 3 --depth 3 --crater split --seed 20261017 \
    --trials 100 --per-level 10 --controls 3 --out OUT_B
./target/release/volcano-rho --bits 48 --ell 2 --depth 3 --max-cofactor 4096 --seed 20261018 \
    --trials 100 --per-level 16 --controls 3 --out OUT_C
./target/release/volcano-rho --bits 48 --ell 2 --depth 3 --max-cofactor 4096 --seed 20261018 \
    --trials 20 --per-level 16 --controls 3 --walk pullback --out OUT_PULLBACK
python plot.py OUT     # matplotlib, plotting only
```

The curve, volcano, instance and every solve's iteration count are a deterministic function of `--seed`; wall times are not. Run from the then-uncommitted `src/bin/volcano_rho.rs` on top of `9b09d308e` (committed in this PR with formatting and lint-only changes), Intel Xeon @ 2.80 GHz, 4 threads, 342.5 s for the rho phase.

Curve B ran on the committed binary (with `--crater`), on the same host, in 754.0 s for the rho phase.

Curve C ran on the committed binary (with the `p ≡ 1 mod 8` draw for ℓ = 2) in 310.7 s for the rho phase. Replaying its first five seeds reproduces 165/165 iteration counts, and curve A still replays 160/160 on this binary.

Replay. The committed binary (formatting and lint fixes only) re-run with `--trials 5` reproduces 160/160 iteration counts of the first five seeds of every arm exactly.

Report PDF: `report.pdf`.